A1S2 EXECUTION CARD

SDS188 · 29 AUGUST 2026 · CHAPTERS 6-9

EMSLEARN NUMERICAL ENTRY

- exactly four decimal places
- full stop decimal separator
- no units/currency symbols unless specifically requested
- preserve full precision during calculation; round final answer only

EXCEL DISCIPLINE

- select method before calculating
- supplied population σ → do NOT replace with STDEV.S
- unknown σ from sample → STDEV.S
- identify X vs $\bar{X}$ vs p before choosing SE
- H_1 determines the hypothesis-test tail
- if a later part explicitly supplies fresh assumed values, use those values for that part rather than automatically carrying prior calculated values forward

FINAL CHECK

☐ correct worksheet ☐ correct object ☐ correct Z/t lane ☐ correct tail
☐ correct critical/p-value route ☐ 4 decimals ☐ no units ☐ decimal point

PRE-ASSESSMENT STUDY RESOURCE - not a live-assessment communication aid.

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Continuous variables + probability as area

1 RECOGNISE IT

- Measurement can take any value on an interval.
- Asked for an interval, tail, density or area.
- A single exact value has probability zero.

2 DECISION / METHOD

1. Name the random variable and its support.
2. Sketch/mark the requested region.
3. Use the distribution CDF or geometric area.
4. Check the answer is between 0 and 1.

3 FORMULA

```
P(a < X < b) = F(b) - F(a)
P(X > a) = 1 - F(a)
P(X = a) = 0
total area = 1
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: uniform and exponential densities, means and variances.
Still know: probability is area; density height is not automatically probability; use complements/differences.

5 EXCEL ROUTE

No universal continuous-probability function. Use the correct family CDF, then:
- left = CDF
- right = 1-CDF
- between = CDF(upper)-CDF(lower)

6 TINY WORKED EXAMPLE

A density is 0.20 on 0 to 5. Then `P(1<X<3)=width x height=2(0.20)=0.40`.

7 TOP EXAM TRAPS

- Adding point probabilities as if X were discrete.
- Treating f(x) as P(X=x).
- Forgetting the support: area outside it is zero.

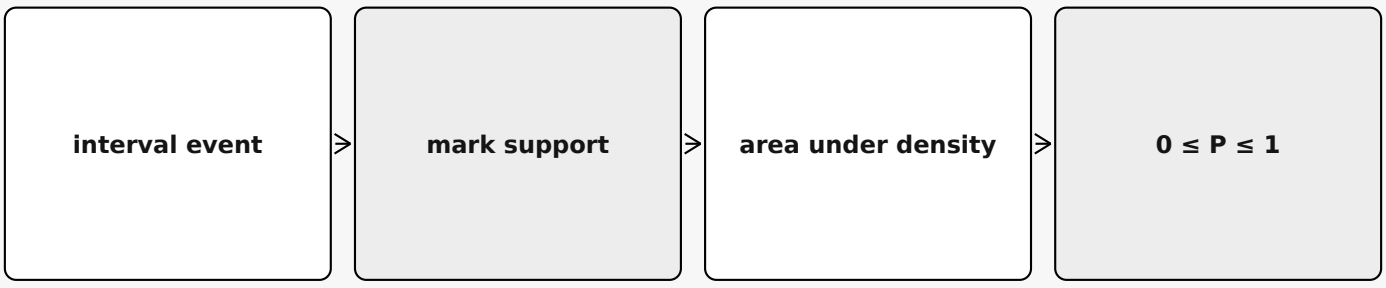
8 SAY ALOUD

For a continuous variable, probability is area over an interval, and one exact point has probability zero.

9 EXIT CHECK

Can I turn a right-tail or between event into CDF areas without looking?

ONE-GLANCE VISUAL ROUTE



Normal distribution + Z standardisation

A1S2 PRE-ASSESSMENT STUDY

1 RECOGNISE IT

- Bell-shaped, symmetric data with mean μ and SD σ.
- Course model notation is $X \sim N(\mu, \sigma^2)$.
- Need to move between original X units and standard Z units.

2 DECISION / METHOD

1. Verify X is modelled as $X \sim N(\mu, \sigma^2)$.
2. Standardise x to z using σ.
3. Use the sign as a reasonableness check.
4. Back-transform only if the answer must be in X units.

3 FORMULA

$X \sim N(\mu, \sigma^2)$
$z = (x - \mu)/\sigma$
$x = \mu + z\sigma$
$Z \sim N(0,1)$

4 FORMULA SHEET VS WHAT I KNOW

Gives: the inference table uses Z when population σ is known.
Still know: raw-score standardisation, back-transformation, and that positive z is above the mean.

5 EXCEL ROUTE

- Direct: `=NORM.DIST(x,mu,sigma,TRUE)`
- Standard: `=NORM.S.DIST(z,TRUE)`
- Do not enter variance where Excel asks for SD.

6 TINY WORKED EXAMPLE

If μ=72, σ=8 and x=84, then z=(84-72)/8=1.5: x is 1.5 SD above the mean.

7 TOP EXAM TRAPS

- COURSE NOTATION uses σ^2 as the second Normal parameter.
- EXCEL NORM.DIST asks for σ, not σ^2.
- Using SE=σ/√n for one observation X.
- Losing the sign of z.

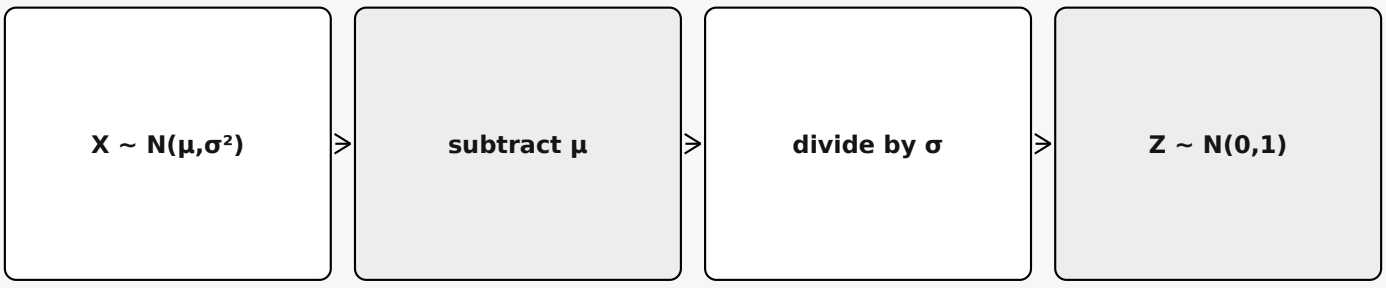
8 SAY ALOUD

Z tells me how many standard deviations the observation is from the population mean.

9 EXIT CHECK

For x below μ, what sign must z have, and why?

ONE-GLANCE VISUAL ROUTE



Normal left, right and between probabilities

1 RECOGNISE IT

- Wording says below/less than, above/more than, or between.
- X is normal and μ, σ are supplied.
- The required answer is an area/probability, not a percentile value.

2 DECISION / METHOD

1. Draw the tail(s).
2. Convert boundaries to z or use NORM.DIST directly.
3. Left: CDF. Right: 1-CDF. Between: upper CDF minus lower CDF.
4. Check area direction visually.

3 FORMULA

`Left: P(X<x)=F(x)`
`Right: P(X>x)=1-F(x)`
`Between: P(a<X<b)=F(b)-F(a)`

4 FORMULA SHEET VS WHAT I KNOW

Gives: no normal tail formulas on the one-page sheet.
Still know: CDF direction, complement, and upper-minus-lower ordering.

5 EXCEL ROUTE

- `=NORM.DIST(x,mu,sigma,TRUE)`
- right: `=1-NORM.DIST(x,mu,sigma,TRUE)`
- between: two NORM.DIST calls, upper minus lower.

6 TINY WORKED EXAMPLE

For $X \sim N(50,10^2)$, `P(40<X<60)=NORM.DIST(60,50,10,TRUE)-NORM.DIST(40,50,10,TRUE)=0.6827`.

7 TOP EXAM TRAPS

- Right tail without the 1-minus.
- Between probabilities added instead of subtracted.
- Confusing X with $\bar{X}$; $\bar{X}$ would use SE.
- Course notation uses variance σ^2, while NORM.DIST uses SD σ.

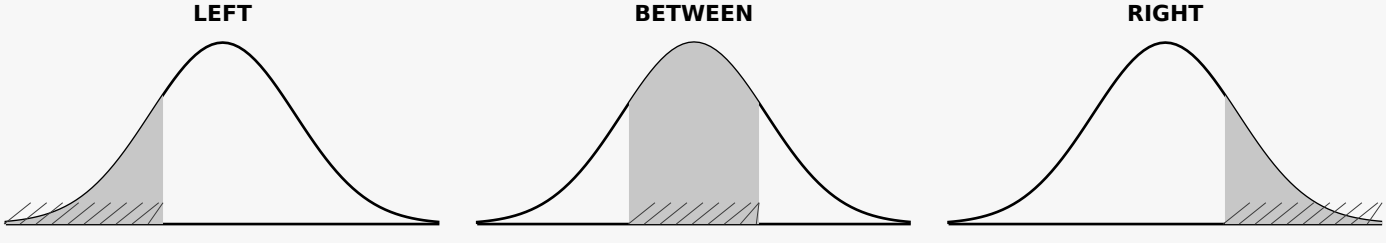
8 SAY ALOUD

Below is the CDF, above is one minus the CDF, and between is upper CDF minus lower CDF.

9 EXIT CHECK

What Excel expression gives $P(X>75)$ for $X \sim N(60,8^2)$?

ONE-GLANCE VISUAL ROUTE



Inverse normal / percentiles

1 RECOGNISE IT

- A cumulative percentage/probability is given and the unknown is x.
- Wording: percentile, cutoff, bottom q%, top q%, central C%.
- This is area -> value, not value -> area.

2 DECISION / METHOD

1. Convert wording to cumulative-left area.
2. Get z or x with an inverse function.
3. If using z, back-transform to X.
4. Check tail and sign.

3 FORMULA

```
z_q = Φ^(-1)(q)
x_q = μ + z_q*σ
top q% uses left area 1-q
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: no inverse-normal formula/function.
Still know: convert all wording to left area and back-transform from z.

5 EXCEL ROUTE

- Direct: `=NORM.INV(q,mu,sigma)`
- Via z: `=NORM.S.INV(q)`, then μ+z*σ.

6 TINY WORKED EXAMPLE

For $X \sim N(100,15^2)$, the 90th percentile is `NORM.INV(0.90,100,15)=119.22`.

7 TOP EXAM TRAPS

- Using 0.10 for the top 10% cutoff instead of 0.90.
- Using NORM.S.DIST instead of NORM.S.INV.
- Reporting z when x units are requested.
- NORM.INV asks for σ, not σ^2.

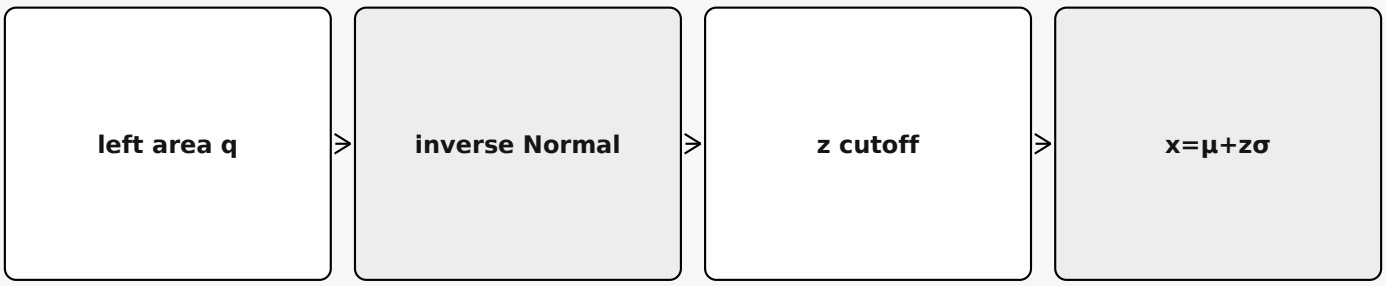
8 SAY ALOUD

Inverse normal starts with a left-tail area and returns the boundary value.

9 EXIT CHECK

What cumulative-left area represents the cutoff for the highest 5%?

ONE-GLANCE VISUAL ROUTE



Evaluating normality + normal probability plot

1 RECOGNISE IT

- Asked whether a normal model/t method is plausible.
- Raw data, histogram, boxplot or normal probability plot is supplied.
- Need an evidence-based normality judgement.

2 DECISION / METHOD

1. Check shape/symmetry and outliers.
2. Compare mean and median; skewness/kurtosis near 0 supports normality.
3. Build/read normal probability plot.
4. Approximately straight positive line -> approximately normal.

3 FORMULA

```
ordered x_(j)
alpha_j = j/(n+1)
z_j = Φ^(-1)(alpha_j)
plot x-axis z_j, y-axis ordered x_(j)
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: nothing for normality plotting.
Still know: straight-line criterion, axes, probability positions, and that one check alone is not proof.

5 EXCEL ROUTE

- Sort X ascending.
- `n=COUNT(range)`; j=1,...,n.
- `α=j/(n+1)`.
- `=NORM.S.INV(alpha)`.
- Insert XY scatter: x=Z, y=ordered X.

6 TINY WORKED EXAMPLE

With n=3 and ordered X={4,5,7}, α values are .25,.50,.75 and Z values about -0.674,0,0.674. Plot (Z,X) and inspect linearity.

7 TOP EXAM TRAPS

- Using NORM.S.DIST(α,TRUE) instead of inverse.
- Swapping axes.
- Calling any upward trend normal despite strong curvature/outliers.

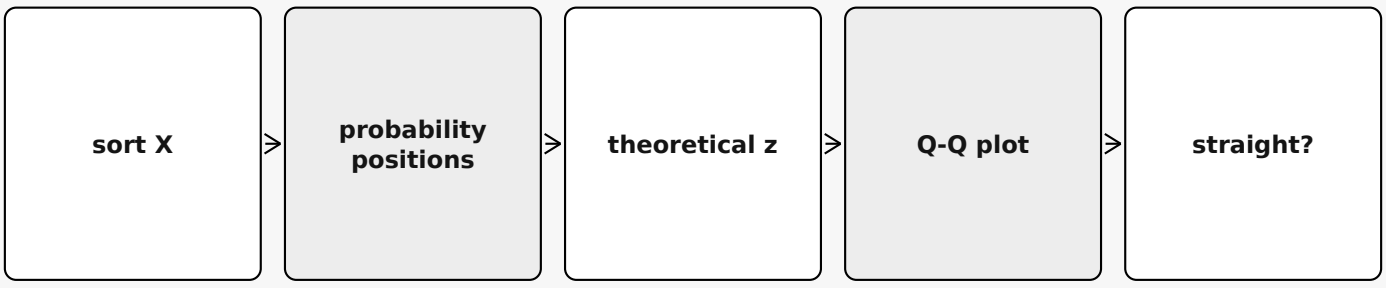
8 SAY ALOUD

Normal data make an approximately straight, positive-slope normal probability plot.

9 EXIT CHECK

Which Excel function converts alpha_j into the theoretical z_j?

ONE-GLANCE VISUAL ROUTE



Uniform distribution

1 RECOGNISE IT

- Every value between fixed bounds a and b is equally likely.
- Rectangular/constant density; zero outside [a,b].
- Probability depends only on interval length.

2 DECISION / METHOD

1. Identify support [a,b].
2. Clip the requested interval to the support.
3. Probability = overlap length / total length.
4. Use midpoint for the mean.

3 FORMULA

$f(x)=1/(b-a),\ a<=x<=b$
$P(c<X<d)=(d-c)/(b-a)$
$E(X)=(a+b)/2$
$Var(X)=(b-a)^2/12$

4 FORMULA SHEET VS WHAT I KNOW

Gives: density, expected value and variance.
Still know: interval probability as length ratio and support clipping.

5 EXCEL ROUTE

No special function needed:
`=(upper-lower)/(b-a)`
SD if needed: `=SQRT((b-a)^2/12)`.

6 TINY WORKED EXAMPLE

For X~Uniform(2,10),
$P(5<X<8)=(8-5)/(10-2)=0.375$.

7 TOP EXAM TRAPS

- Saying every interval has equal probability; only equal-length intervals do.
- Using density 1/(b-a) as the final interval probability.
- Including area outside [a,b].

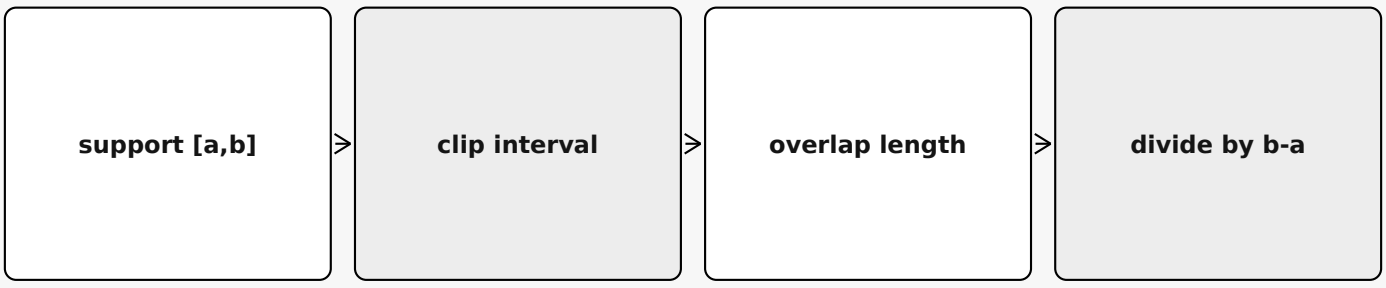
8 SAY ALOUD

Uniform probability is the requested overlap length divided by the full support length.

9 EXIT CHECK

For Uniform(-2,6), what is P(X>4)?

ONE-GLANCE VISUAL ROUTE



Exponential distribution + inverse exponential

1 RECOGNISE IT

- Waiting time/distance between random events occurring at average rate lambda.
- Right-skewed, support x>0.
- Rate and time units must match.

2 DECISION / METHOD

1. Convert rate and time to the same unit.
2. Left wait probability uses 1-exp(-lambda*t).
3. Right wait probability uses exp(-lambda*t).
4. For a percentile, solve the CDF for t.

3 FORMULA

```
f(x)=lambda*exp(-lambda*x)
F(t)=1-exp(-lambda*t)
P(X>t)=exp(-lambda*t)
t_q=-LN(1-q)/lambda
E(X)=1/lambda; Var=1/lambda^2
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: density, mean and variance.
Still know: left/right CDFs, inverse formula, memoryless context, and unit conversion.

5 EXCEL ROUTE

- Left: `=EXPON.DIST(t,lambda,TRUE)`
- Right: `=1-EXPON.DIST(t,lambda,TRUE)`
- Inverse: `=-LN(1-q)/lambda`.

6 TINY WORKED EXAMPLE

At lambda=4 per hour, `P(wait>0.25h)=EXP(-4*0.25)=0.3679`.

7 TOP EXAM TRAPS

- Mixing minutes with a per-hour rate.
- Using 1-exp(...) for a right tail.
- Treating lambda as the mean wait; mean wait is 1/lambda.

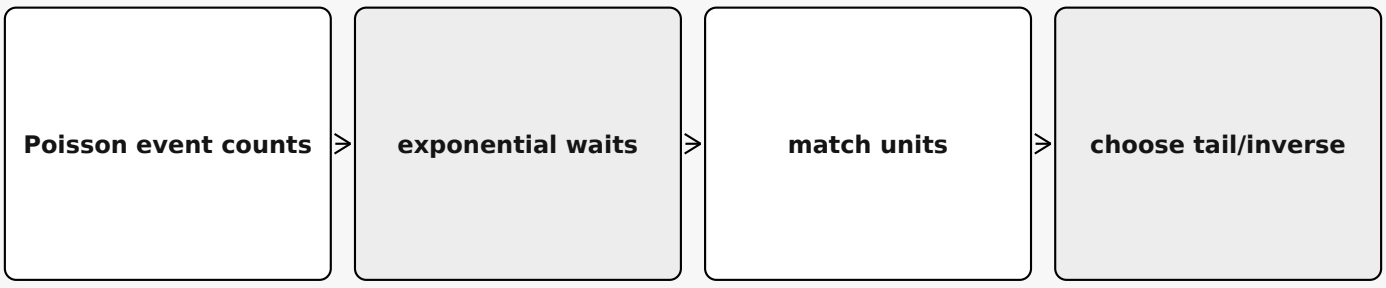
8 SAY ALOUD

Exponential waiting-time questions live or die on matching units and choosing left versus right tail.

9 EXIT CHECK

If F(t)=0.80, what equation gives the 80th-percentile wait?

ONE-GLANCE VISUAL ROUTE



Normal approximation to binomial + continuity correction

1 RECOGNISE IT

- X is a binomial count with large n.
- Exact binomial calculation is replaced by a normal model.
- Required course checks: nπ > 5 AND n(1-π) > 5.

2 DECISION / METHOD

1. Confirm nπ > 5 and n(1-π) > 5.
2. Set mean=nπ and SD=√[nπ(1-π)].
3. Apply the correct ±0.5 continuity correction.
4. Use the normal CDF/tail.

3 FORMULA

$Y \approx N(n\pi, n\pi(1-\pi))$
$SD=\sqrt{[n\pi(1-\pi)]}$
$X \le k \rightarrow Y < k+0.5$
$X \ge k \rightarrow Y > k-0.5$
$a \le X \le b \rightarrow a-0.5 < Y < b+0.5$

4 FORMULA SHEET VS WHAT I KNOW

Gives: no normal-approximation or continuity-correction rule.
Still know: conditions, normal mean/variance, and boundary shifts.

5 EXCEL ROUTE

After calculating mean and SD:
`=NORM.DIST(k+0.5,mean,SD,TRUE)`
Right tail uses 1 minus the appropriate CDF.

6 TINY WORKED EXAMPLE

If n=100, π=.40, then mean=40 and SD=sqrt(24). For P(X<=45), use `NORM.DIST(45.5,40,SQRT(24),TRUE)`.

7 TOP EXAM TRAPS

- No continuity correction.
- Wrong direction: X>k uses k+0.5, but X>=k uses k-0.5.
- Using variance where Excel wants SD.

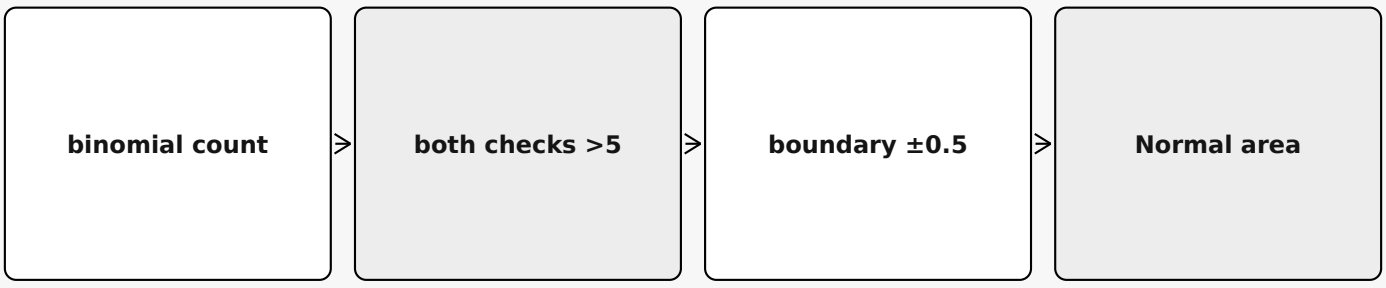
8 SAY ALOUD

Approximate the binomial with mean n*π, SD sqrt(n*π(1-π)), then move the integer boundary by half a unit.

9 EXIT CHECK

What continuous interval represents 12<=X<=18?

ONE-GLANCE VISUAL ROUTE



Population -> sample -> statistic -> sampling distribution

1 RECOGNISE IT

- Repeated random samples of the same size are imagined.
- The random object is a statistic such as $\bar{X}$ or p.
- Asked about how a statistic varies across samples.

2 DECISION / METHOD

Population (μ,σ,π) -> draw sample of size n -> compute statistic ($\bar{X}$,p) -> repeat conceptually -> distribution of statistic.

3 FORMULA

population variable: X
sample statistic: $\bar{X}$ or p
sampling distribution: all possible statistic values for fixed n

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter/statistic pairings and their standard errors under conditions.
Still know: the four-stage chain and which random object the probability concerns.

5 EXCEL ROUTE

Use Excel only after naming the object:
- raw sample mean: =AVERAGE(range)
- sample size: =COUNT(range)
- probability model comes from the sampling distribution.

6 TINY WORKED EXAMPLE

Population values generate many samples. If each sample has n=25, every sample gives a different $\bar{X}$; those $\bar{X}$ values form the sampling distribution.

7 TOP EXAM TRAPS

- Calling the sample histogram the sampling distribution.
- Using raw-X SD when the event is about $\bar{X}$.
- Confusing parameter μ with statistic $\bar{X}$.

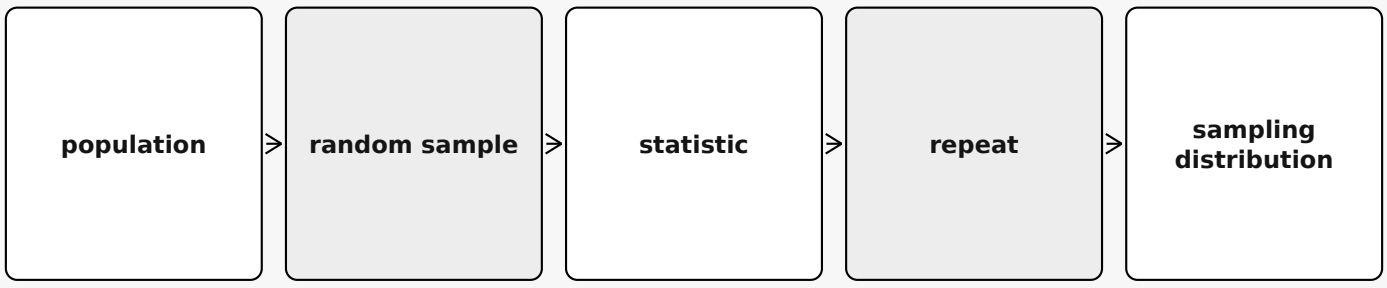
8 SAY ALOUD

A sampling distribution is the distribution of a statistic over all possible samples of one fixed size.

9 EXIT CHECK

In $P(\bar{X}>10)$, what is random: one observation or the sample mean?

ONE-GLANCE VISUAL ROUTE



Unbiasedness + exact sampling distributions

1 RECOGNISE IT

- Small finite population; asked to list all possible samples/statistics.
- Sampling rule and replacement status are specified.
- Asked whether an estimator is unbiased.

2 DECISION / METHOD

1. List all allowed samples under the stated sampling rule.
2. Compute the statistic for each sample.
3. Build its probability/frequency distribution.
4. Compare the mean of that distribution with the parameter.

3 FORMULA

`unbiased if E(estimator)=parameter`

$E(\bar{X})=\mu$

$E(p)=\pi$

4 FORMULA SHEET VS WHAT I KNOW

Gives: the inference table pairs $\bar{X}$ with μ and p with π.

Still know: exact enumeration, probabilities, and the expected-value definition of unbiasedness.

5 EXCEL ROUTE

For small enumeration tables:

- use rows for allowed samples
- `=AVERAGE(sample_cells)`
- average the full statistic column to check the centre.

6 TINY WORKED EXAMPLE

Population {2,4}; n=1 samples give $\bar{X}$ values 2 and 4 equally. Their mean is 3, equal to population $\mu=3$, so $\bar{X}$ is unbiased.

7 TOP EXAM TRAPS

- Unbiased does not mean every estimate equals the parameter.
- Ignoring replacement/order rules.
- Checking only one sample instead of E(estimator).

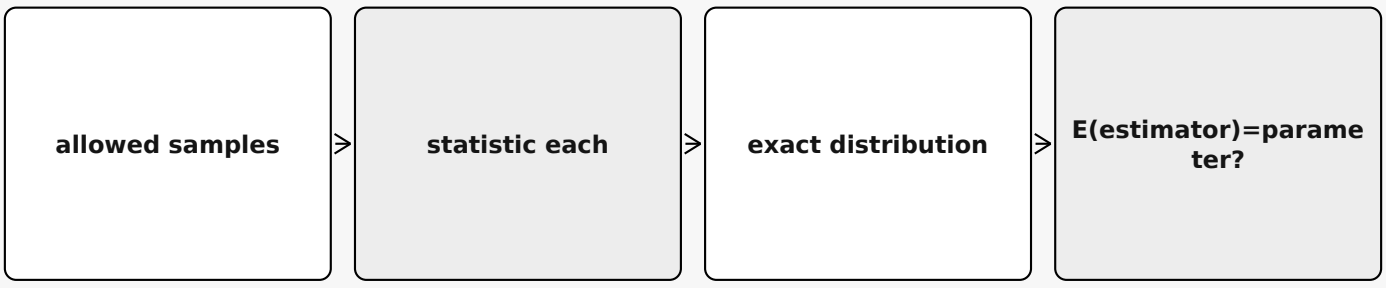
8 SAY ALOUD

Unbiased means the estimator is centred correctly over repeated samples, not that every sample is correct.

9 EXIT CHECK

What equality must hold for $\bar{X}$ to be an unbiased estimator of μ?

ONE-GLANCE VISUAL ROUTE



Standard error + X versus X̄

1 RECOGNISE IT

- Compare one observation with an average of n observations.
- Asked for the spread of sample means or effect of larger n.
- Population σ is known.

2 DECISION / METHOD

1. Name the random object: X or X̄.
2. X uses centre μ and SD σ.
3. X̄ uses centre μ and SE σ/√n.
4. Larger n narrows X̄, not the population itself.

3 FORMULA

$SD(X)=\sigma$
$SE(\bar{X})=\sigma/\sqrt{n}$
$Var(\bar{X})=\sigma^2/n$
$E(\bar{X})=\mu$

4 FORMULA SHEET VS WHAT I KNOW

Gives: SE(X̄)=σ/√n.
Still know: SD describes individuals; SE describes statistic-to-statistic variability.

5 EXCEL ROUTE

- `=sigma/SQRT(n)`
- raw sample: `=AVERAGE(range)`
- never use STDEV.S when population σ is explicitly supplied.

6 TINY WORKED EXAMPLE

If σ=18 and n=81, `SE(Xbar)=18/SQRT(81)=2`. Individuals still have SD 18.

7 TOP EXAM TRAPS

- Using σ instead of SE for X̄.
- Dividing by n instead of √n.
- Saying larger n reduces population SD.

8 SAY ALOUD

X varies with σ; sample means vary with σ divided by the square root of n.

9 EXIT CHECK

If n is multiplied by 4, what happens to SE(X̄)?

ONE-GLANCE VISUAL ROUTE



Sample mean, σ known -> Z

1 RECOGNISE IT

- Probability concerns $\bar{X}$, not one X.
- Population σ is explicitly known.
- Population normal, or CLT justifies approximate normality.

2 DECISION / METHOD

1. Check the $\bar{X}$ normality condition.
2. Compute SE=σ/√n.
3. Standardise the $\bar{X}$ boundary.
4. Use left/right/between normal area.

3 FORMULA

$\bar{X} \sim N(\mu, \sigma^2/n)$
$z=(\bar{x}-\mu)/(\sigma/\sqrt{n})$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $\bar{X}$, SE=σ/√n, Z when σ known.
Still know: the event direction and probability conversion.

5 EXCEL ROUTE

Direct left tail:
`=NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE)`
Right = 1 minus; between = upper minus lower.

6 TINY WORKED EXAMPLE

μ=50, σ=12, n=36. For P($\bar{X}$<53), SE=2 and z=1.5, so probability is about 0.9332.

7 TOP EXAM TRAPS

- Using σ instead of σ/√n.
- Using t despite known population σ.
- Applying a normal model without population normality or adequate n.

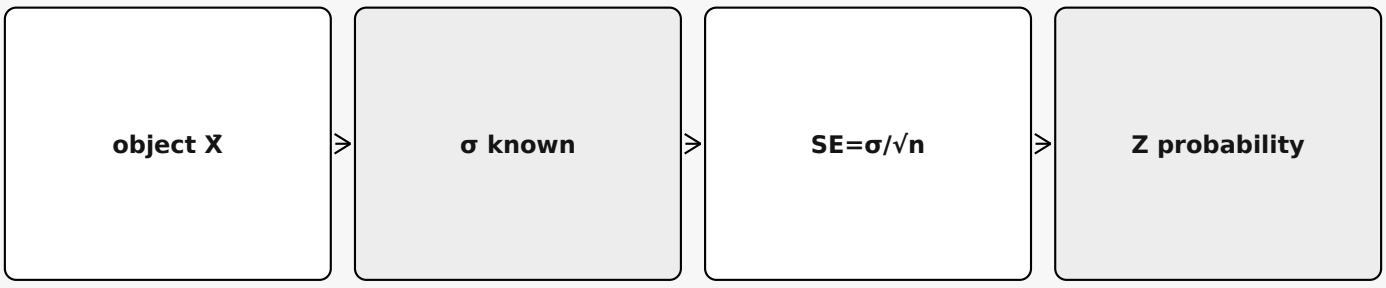
8 SAY ALOUD

Known population σ means the sample mean uses a Z model with standard error σ over square root n.

9 EXIT CHECK

What Excel formula gives P($\bar{X}$>53) for the tiny example?

ONE-GLANCE VISUAL ROUTE



Central Limit Theorem

1 RECOGNISE IT

- Population shape is not normal or is unknown.
- Probability concerns a sample mean.
- Asked whether $\bar{X}$ is approximately normal for a stated n.

2 DECISION / METHOD

1. Identify the population shape.
2. If population normal: $\bar{X}$ normal for any n.
3. Official guide: fairly symmetric often n>15; most distributions n>30.
4. Use centre μ and SE σ/√n.

3 FORMULA

for sufficiently large n:

$\bar{X}$ approx $N(\mu, \sigma^2/n)$

4 FORMULA SHEET VS WHAT I KNOW

Gives: SE formula under conditions.
Still know: CLT concerns the sampling distribution of $\bar{X}$, not the raw population becoming normal.

5 EXCEL ROUTE

Once justified, use:
`=NORM.DIST(boundary,mu,sigma/SQRT(n),TRUE)`
with complements/differences as needed.

6 TINY WORKED EXAMPLE

A skewed population has μ=20, σ=10 and n=100. CLT gives $\bar{X}$ approximately $N(20,1^2)$.

7 TOP EXAM TRAPS

- Claiming raw observations become normal.
- Ignoring strong skew with small n.
- Forgetting that SE shrinks as n grows.

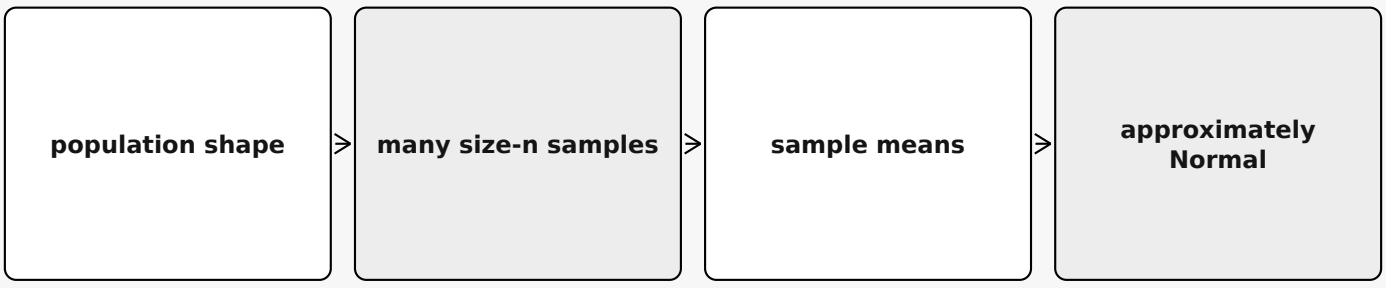
8 SAY ALOUD

The CLT makes sample means approximately normal for large n, while keeping centre μ and spread σ/√n.

9 EXIT CHECK

Does the CLT make the original population normal?

ONE-GLANCE VISUAL ROUTE



Sample mean, σ unknown -> Student t

1 RECOGNISE IT

- Probability concerns $\bar{X}$.
- Population σ is unknown and sample s is supplied.
- Population is normal/approximately normal under the course conditions.

2 DECISION / METHOD

1. Use s/√n as estimated SE.
2. Set df=n-1.
3. Convert boundaries to t.
4. Use T.DIST and the correct tail/difference.

3 FORMULA

$t=(\bar{x}-\mu)/(s/\sqrt{n})$
$df=n-1$

4 FORMULA SHEET VS WHAT I KNOW

Gives: s/√n, t_(n-1) when σ is unknown.
Still know: normality condition, df, and tail handling.

5 EXCEL ROUTE

- Left: `=T.DIST(tcalc,df,TRUE)`
- Right: `=1-T.DIST(tcalc,df,TRUE)`
- Between: upper T.DIST minus lower T.DIST.

6 TINY WORKED EXAMPLE

μ=30, s=5, n=25. For P($\bar{X}$<32), `t=(32-30)/(5/SQRT(25))`=2, df=24; use T.DIST(2,24,TRUE).

7 TOP EXAM TRAPS

- Using Z because n looks large while σ is unknown in this course lane.
- df=n.
- Using σ notation for sample s.

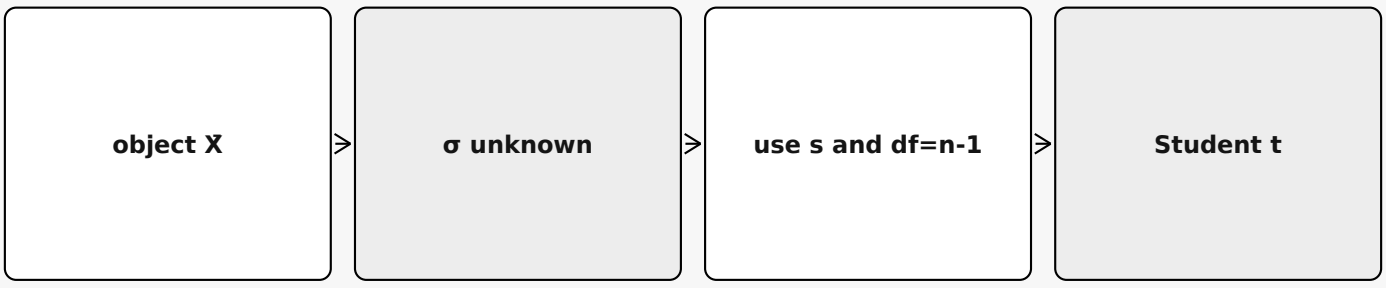
8 SAY ALOUD

Unknown σ means estimate the standard error with s and use Student t with n-1 degrees of freedom.

9 EXIT CHECK

Which Excel function gives a cumulative-left t probability?

ONE-GLANCE VISUAL ROUTE



Sampling distribution of a proportion

1 RECOGNISE IT

- Each observation is success/failure.
- The random statistic is p=X/n.
- Population proportion π is known for the probability model.

2 DECISION / METHOD

1. Check nπ > 5 AND n(1-π) > 5.
2. Centre at π.
3. Use theoretical SE with π.
4. Standardise the p boundary and use normal areas.

3 FORMULA

p=X/n
E(p)=π
SE(p)=sqrt(π*(1-π)/n)
z=(p-π)/SE(p)

4 FORMULA SHEET VS WHAT I KNOW

Gives: p estimates π; theoretical SE uses π; Z model.
Still know: approximation conditions and p versus π roles.

5 EXCEL ROUTE

Direct left tail:
=NORM.DIST(p_boundary,pi,SQRT(pi*(1-pi)/n),TRUE)
Use 1-minus/right or upper-minus-lower.

6 TINY WORKED EXAMPLE

π=.40, n=100. For P(p<.50), SE=.0490 and z=2.041; probability is about .9794.

7 TOP EXAM TRAPS

- Weakening the strict sourced course inequality.
- Substituting observed p into this theoretical probability SE.
- Treating p as a count.
- Forgetting that π is fixed and p is random.

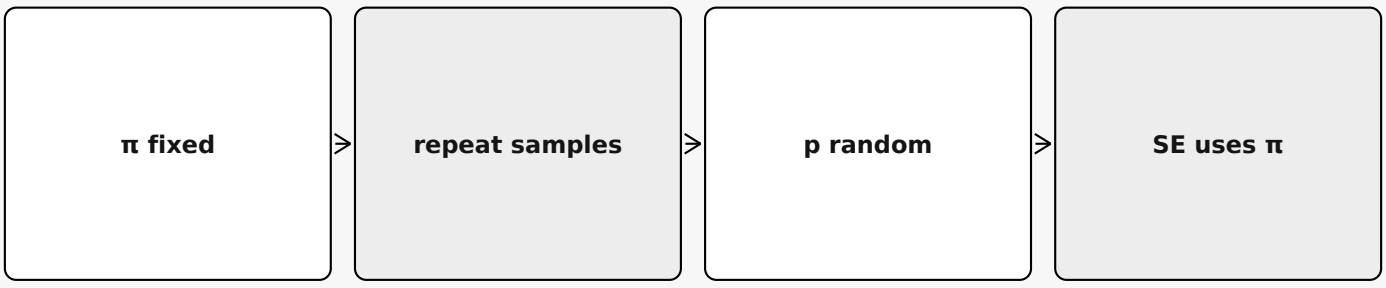
8 SAY ALOUD

For sampling probabilities, p is random, π is the centre, and theoretical SE uses π.

9 EXIT CHECK

Which symbol belongs in SE(p) for a sampling-distribution probability: p or π?

ONE-GLANCE VISUAL ROUTE



Difference between two independent sample means

1 RECOGNISE IT

- Two independent random samples are compared.
- Event concerns $\bar{X}_1-\bar{X}_2$.
- Population sigmas are known; populations normal or both n values are large.

2 DECISION / METHOD

1. Fix order: group 1 minus group 2.
2. Centre at $\mu_1-\mu_2$.
3. Combine variances, not SDs.
4. Standardise the difference and use normal areas.

3 FORMULA

$E(\bar{X}_1-\bar{X}_2)=\mu_1-\mu_2$

$SE=\sqrt{\sigma_1^2/n1+\sigma_2^2/n2}$

$z=[(d)-(\mu_1-\mu_2)]/SE$

4 FORMULA SHEET VS WHAT I KNOW

Gives: statistic, parameter difference, combined SE and Z when both sigmas are known.
Still know: independence, fixed subtraction order, and event direction.

5 EXCEL ROUTE

For $P(\bar{X}_1-\bar{X}_2>c)$:

```
=1-NORM.DIST(c,mu1-mu2,SQRT(sigma1^2/n1+sigma2^2/n2),TRUE)
```

6 TINY WORKED EXAMPLE

$\mu_1-\mu_2=5$, $\sigma_1=3$,n1=25,$\sigma_2=4$,n2=16. SE=sqrt(1.36)=1.166. P(diff>7) uses z=1.715, about .0432.

7 TOP EXAM TRAPS

- Reversing group order mid-solution.
- Adding SDs instead of variances.
- Using paired methods for independent samples.

8 SAY ALOUD

For independent means, subtract in one fixed order and add their sampling variances under the square root.

9 EXIT CHECK

What is the centre of $\bar{X}_1-\bar{X}_2$?

ONE-GLANCE VISUAL ROUTE

LOCK: group 1-group 2 → **add variances** → **divide by SE** → **Normal area**

Finite-population correction (FPC)

⚠ PROVISIONAL / LEGACY-SUPPORTED BRIDGE

1 RECOGNISE IT

- A finite population size N is supplied.
- Sampling is without replacement.
- Sampling fraction n/N is more than 0.05.

2 DECISION / METHOD

1. Confirm finite N and without replacement.
2. Compute ordinary SE.
3. Compute FPC.
4. Multiply SE by FPC before standardising.
5. If with replacement or n/N<=0.05, keep ordinary SE.

3 FORMULA

```
FPC=sqrt((N-n)/(N-1))
adjusted SE = ordinary SE * FPC
```

4 FORMULA SHEET VS WHAT I KNOW

Status: C7.52-C7.58 retain the canonical master's provisional scope safeguard. Bridge: retained as a sourced bridge, especially into Chapter 8; not silently promoted as current Chapter 7 core.

5 EXCEL ROUTE

- =SQRT((N-n)/(N-1))
- adjusted mean SE: =sigma/SQRT(n)*FPC
- adjusted p SE: theoretical SE times FPC.

6 TINY WORKED EXAMPLE

N=500,n=50 without replacement: FPC=SQRT(450/499)=0.9496. Multiply the ordinary SE by 0.9496.

7 TOP EXAM TRAPS

- Applying FPC merely because N is stated.
- Applying it when sampling with replacement.
- Omitting the square root.
- Failing to inspect n/N.

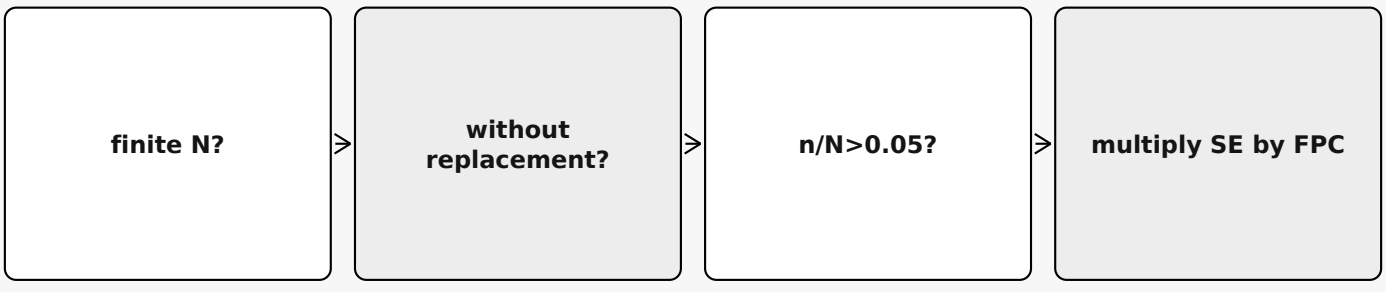
8 SAY ALOUD

Use FPC only for material sampling without replacement from a finite population: inspect replacement and n/N first.

9 EXIT CHECK

N=1000,n=20 without replacement: does the 5% rule trigger FPC?

ONE-GLANCE VISUAL ROUTE



Point vs interval estimation + estimator characteristics

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Asked to distinguish one estimate from a range.
- Need to judge an estimator by repeated-sampling behaviour.
- Parameter is unknown; statistic estimates it.

2 DECISION / METHOD

1. Match parameter to statistic: μ <- X̄; π <- p.
2. Point estimate = one statistic value.
3. Interval estimate = plausible range plus confidence level.
4. Judge unbiasedness, consistency and relative efficiency.

3 FORMULA

```
unbiased: E(estimator)=parameter
consistent: error shrinks as n grows
more efficient: smaller variance
among unbiased estimators
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter/statistic pairings and standard errors.
Still know: definitions of point, interval, unbiasedness, consistency and efficiency.

5 EXCEL ROUTE

- Mean point estimate: `=AVERAGE(range)`
- n: `=COUNT(range)`
- Proportion point estimate: successes/count.

6 TINY WORKED EXAMPLE

A sample gives X̄=42.6. That is a point estimate of μ. A 95% CI such as (40.1,45.1) adds precision information.

7 TOP EXAM TRAPS

- Saying the parameter changes from sample to sample.
- Unbiased means 'always exactly correct'.
- Calling a confidence level a probability that this fixed interval is correct.

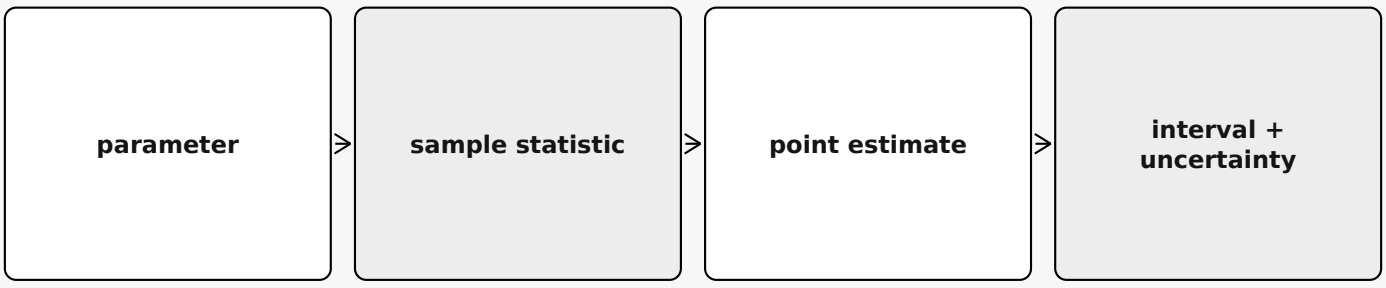
8 SAY ALOUD

The statistic is my sample-based point estimate; the confidence interval shows its uncertainty about the fixed parameter.

9 EXIT CHECK

Which estimator characteristic prefers smaller variance when both estimators are unbiased?

ONE-GLANCE VISUAL ROUTE



Confidence, α, critical value, SE and margin of error

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Any confidence-interval question before selecting a parameter-specific method.
- Confidence level C is supplied.
- Need critical value, SE, margin or endpoints.

2 DECISION / METHOD

1. α=1-C.
2. For a two-sided CI, place α/2 in each tail.
3. Select Z or t.
4. Compute SE.
5. Margin e*=critical×SE; CI=estimate±e*.

3 FORMULA

```
CI = estimate ± critical×SE
α = 1-C
z* = Φ⁻¹(1-α/2) > 0
t* = T.INV.2T(α,df) > 0
margin = critical×SE
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter-specific SE formulas.
Still know: general CI structure, confidence-to-α conversion, and when critical is Z versus t.

5 EXCEL ROUTE

- Z critical: `=NORM.S.INV(1-alpha/2)`
- t critical: `=T.INV.2T(alpha,df)`
- endpoints: estimate ± critical*SE.

6 TINY WORKED EXAMPLE

For C=.90, α=.10 and each tail=.05. The positive Z critical is NORM.S.INV(.95)=1.6449.

7 TOP EXAM TRAPS

- Using α instead of α/2 for a two-sided CI.
- Confusing confidence level with critical value.
- Using SD where SE is required.

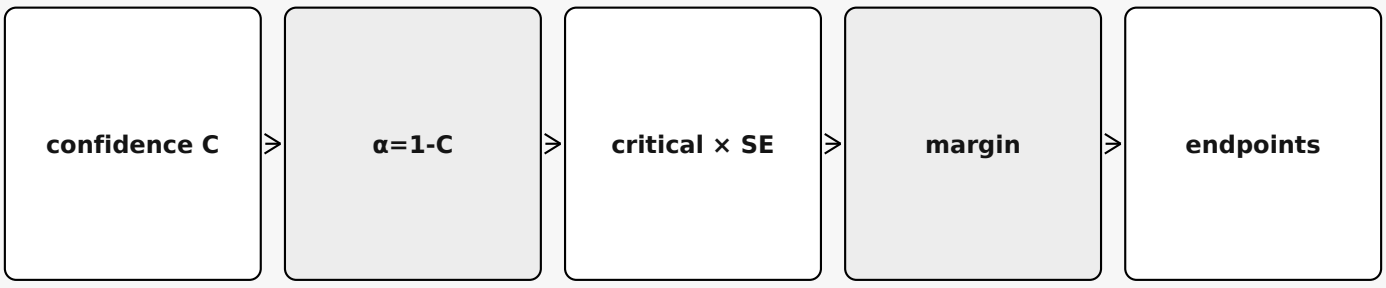
8 SAY ALOUD

Every confidence interval is point estimate plus or minus a critical value times a standard error.

9 EXIT CHECK

For 98% confidence, what cumulative-left area gives the positive Z critical?

ONE-GLANCE VISUAL ROUTE



Mean CI, σ known -> Z

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate population mean μ.
- Population σ is explicitly supplied/known.
- Random sample; population normal or sample large enough.

2 DECISION / METHOD

1. Compute $\bar{X}$ and n.
2. SE=σ/√n.
3. z*=NORM.S.INV(1-α/2) > 0.
4. $\bar{X}$±z*SE.
5. Report units and confidence.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$CI: \bar{X} \pm z^* \times \sigma/\sqrt{n}$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $\bar{X}$, SE=σ/√n, Z lane when σ known. Still know: critical value, CI endpoints and interpretation.

5 EXCEL ROUTE

- `n=COUNT(range); Xbar=AVERAGE(range)`
- `zstar=NORM.S.INV(1-alpha/2)`
- lower/upper = `Xbar +/- zstar*sigma/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=40,σ=6,n=36,C=.95. SE=1 and z=1.96, so CI is `(38.04,41.96)`.

7 TOP EXAM TRAPS

- Using sample s when σ is supplied.
- Using t in the known-σ lane.
- Treating the CI as a range for individual X values.

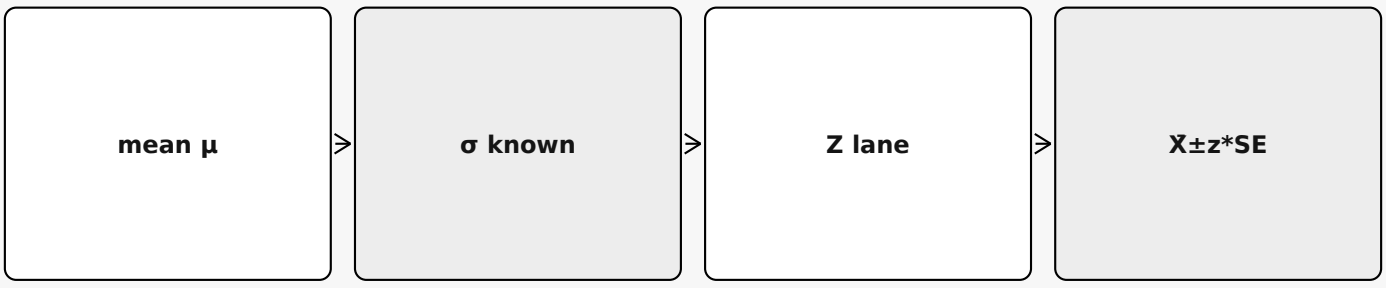
8 SAY ALOUD

Known population σ means a Z interval for μ centred at $\bar{X}$ with SE σ/√n.

9 EXIT CHECK

What changes in the interval if n quadruples but everything else stays fixed?

ONE-GLANCE VISUAL ROUTE



Student t, df and mean CI when σ unknown

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate population mean μ.
- Population σ unknown; sample s available.
- Population approximately normal under the course conditions.

2 DECISION / METHOD

1. Compute n, $\bar{X}$, s.
2. df=n-1.
3. t*=T.INV.2T(α,df).
4. SE=s/√n.
5. $\bar{X}$ ± t*SE; interpret subject to normality.

3 FORMULA

```
t* = T.INV.2T(α,df) > 0
CI: X¯ ± t* × s/√n
df=n-1
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: s/√n and t_(n-1) when σ unknown. Still know: t critical lookup/function, normality condition and endpoint construction.

5 EXCEL ROUTE

- `n=COUNT(range)`; `Xbar=AVERAGE(range)`; `s=STDEV.S(range)`
- `tstar=T.INV.2T(alpha,n-1)`
- endpoints: `Xbar +/- tstar*s/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=25,s=4,n=16,C=.95. df=15,t*=2.131,SE=1 -> CI (22.869,27.131).

7 TOP EXAM TRAPS

- df=n.
- Using Z because n seems moderate.
- Ignoring normality for a small sample.
- Mixing σ and s.

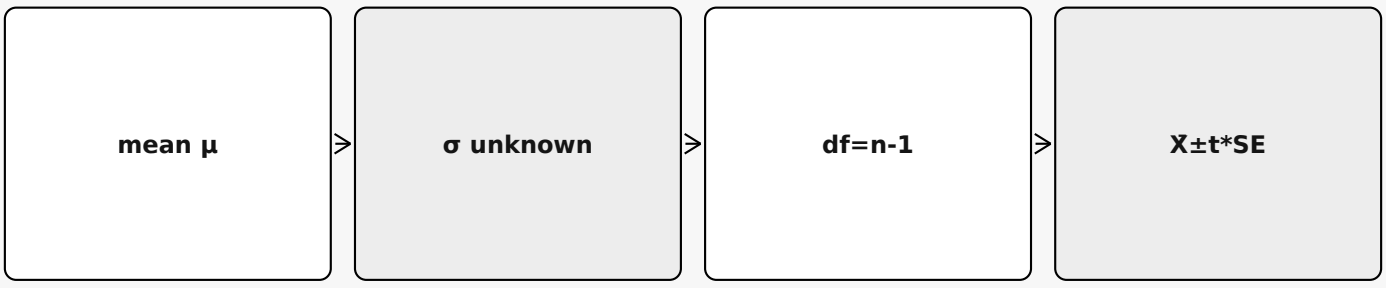
8 SAY ALOUD

Unknown σ means Student t with sample s and n-1 degrees of freedom.

9 EXIT CHECK

Which Excel function returns the positive two-sided t critical value?

ONE-GLANCE VISUAL ROUTE



Population-proportion confidence interval

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate a population percentage/proportion π.
- Sample has x successes out of n.
- Need a two-sided CI and normal approximation is adequate.

2 DECISION / METHOD

1. p=x/n.
2. Check np > 5 AND n(1-p) > 5.
3. Estimated SE=√[p(1-p)/n].
4. CI=p±z*SE.
5. Convert to percent only if requested.

3 FORMULA

p=x/n
SE_hat=√[p(1-p)/n]
$z^* = \Phi^{-1}(1-\alpha/2) > 0$
CI: p ± z*SE_hat

4 FORMULA SHEET VS WHAT I KNOW

Gives: theoretical SE uses π for a sampling distribution.
Still know: a CI estimates unknown π, so substitute sample p into estimated SE.

5 EXCEL ROUTE

- p=successes/n
- SE=SQRT(p*(1-p)/n)
- z=NORM.S.INV(1-alpha/2)
- endpoints p ± z*SE.

6 TINY WORKED EXAMPLE

x=60,n=100,C=.95: p=.60,SE=.0490,margin=.0960 -> CI (.5040,.6960).

7 TOP EXAM TRAPS

- Weakening the strict sourced course inequality.
- Using null π_0 in a CI SE.
- Forgetting p is a proportion, not a percentage.
- Using π instead of sample p in the estimated CI SE.

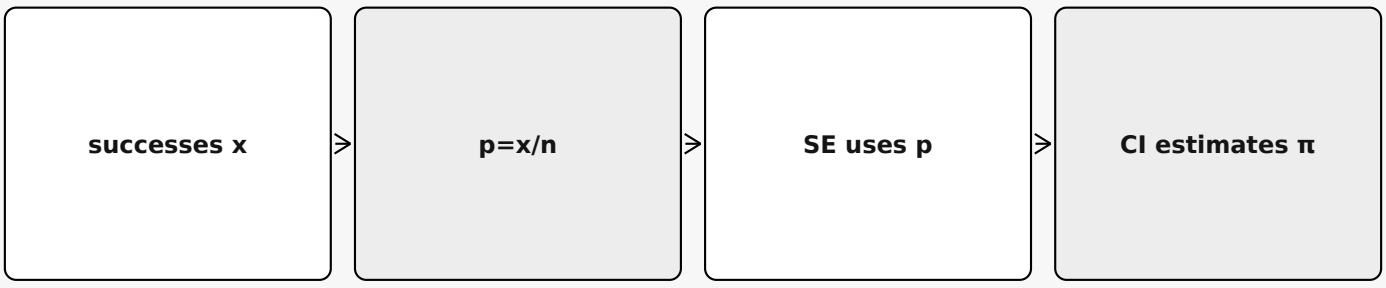
8 SAY ALOUD

A proportion CI uses sample p both as the centre and inside the estimated standard error.

9 EXIT CHECK

Why does a CI for π use p rather than an unknown π inside its estimated SE?

ONE-GLANCE VISUAL ROUTE



Required sample size for a mean

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need n to estimate μ within margin e at confidence C.
- Population σ, pilot s, or conservative expected SD is supplied.
- This is planning for precision, not a completed CI.

2 DECISION / METHOD

1. Convert C to z*.
2. Choose σ estimate.
3. Compute $n0=(z^*\sigma/e)^2$.
4. ALWAYS ROUND UP.
5. Apply finite-population adjustment only if separately triggered.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$n_0 = (z^*\sigma/e)^2$

ALWAYS ROUND UP

4 FORMULA SHEET VS WHAT I KNOW

Gives: required-mean sample-size formula.
Still know: how to get z, estimate σ if unknown, and always round up.

5 EXCEL ROUTE

```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*sigma^2)/e^2,0)
```

6 TINY WORKED EXAMPLE

C=.95,σ=10,e=3:
$n0=(1.96*10/3)^2=42.684$, so required n=43.

7 TOP EXAM TRAPS

- Rounding to nearest/down.
- Using full interval width instead of margin e.
- Using t critical in this planning formula.
- Forgetting FPC is a later step.

8 SAY ALOUD

Mean sample size comes from the margin equation and is always rounded up.

9 EXIT CHECK

If the allowed margin halves, by what factor does required n change?

ONE-GLANCE VISUAL ROUTE



Required sample size for a proportion

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need n to estimate π within margin e at confidence C.
- Prior/pilot proportion is supplied, or none is available.
- Planning for a future proportion estimate.

2 DECISION / METHOD

1. Get z* from confidence.
2. Choose π estimate: pilot/prior; if none, use .50.
3. Compute n0.
4. ALWAYS ROUND UP.
5. Adjust for finite N only if required.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$n_0 = z^{*2}\pi(1-\pi)/e^2$

no estimate → π=.50

ALWAYS ROUND UP

4 FORMULA SHEET VS WHAT I KNOW

Gives: required-proportion sample-size formula.

Still know: .50 is conservative, convert percent margin to decimal, and round up.

5 EXCEL ROUTE

```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*pi*(1-pi))/e^2,0)
```

6 TINY WORKED EXAMPLE

C=.95, prior π=.40,e=.05: n0=368.79, so n=369.

7 TOP EXAM TRAPS

- Using observed x/n when no pilot exists instead of .50.
- Entering e=5 instead of .05.
- Rounding down.
- Confusing CI SE p with planning π.

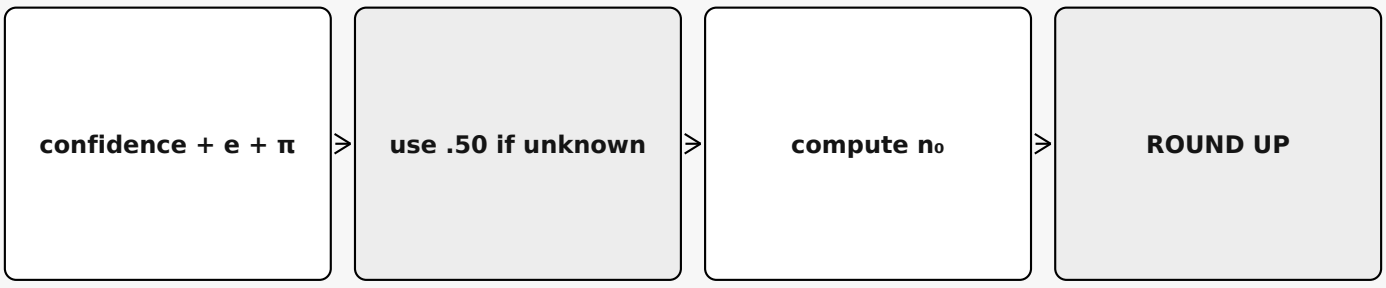
8 SAY ALOUD

For proportion sample size, use the best prior π or conservative .50, then always round up.

9 EXIT CHECK

Why does π=.50 produce the largest planned n?

ONE-GLANCE VISUAL ROUTE



8 Finite-population correction for confidence intervals

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Finite N, sample without replacement, and n/N>0.05.
- Need a CI for μ or π.
- Sampling a material fraction reduces uncertainty.

2 DECISION / METHOD

1. Build the ordinary SE.
2. FPC=sqrt((N-n)/(N-1)).
3. adjusted SE=ordinary SE*FPC.
4. Use the usual Z/t critical and centre.
5. No FPC for replacement or negligible fraction.

3 FORMULA

FPC=√[(N-n)/(N-1)]
mean CI: $\bar{X} \pm$ critical×(s/√n)×FPC
proportion CI: p ± z*√[p(1-p)/n]×FPC

4 FORMULA SHEET VS WHAT I KNOW

Gives: finite-population sample-size adjustment only.
Still know: CI FPC multiplier, trigger, and replacement caveat.

5 EXCEL ROUTE

- FPC=SQRT((N-n)/(N-1))
- multiply ordinary SE by FPC before the margin.
- retain Z/t choice from the underlying CI method.

6 TINY WORKED EXAMPLE

N=1000,n=100,p=.30,C=.90: FPC=.9492; adjusted SE=.0435. Using full precision, margin=.0715 and CI≈(.2285,.3715).

7 TOP EXAM TRAPS

- Applying FPC to the centre.
- Dividing by FPC.
- Applying it when sampling with replacement.
- Forgetting the n/N trigger.

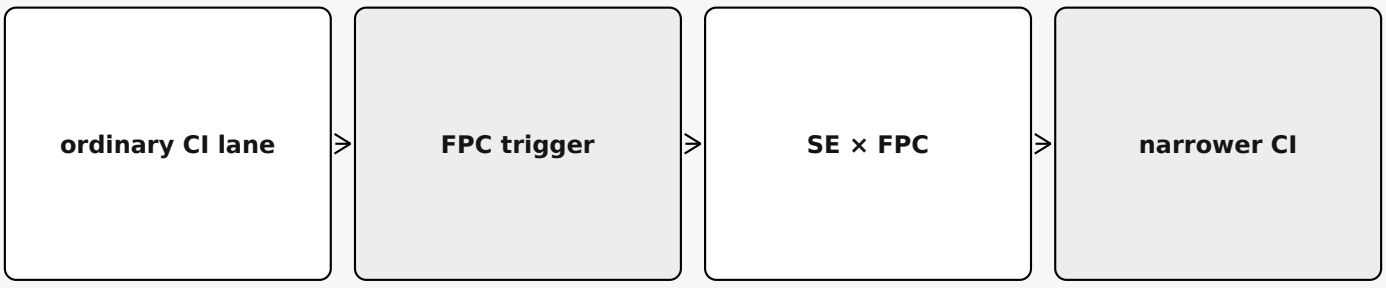
8 SAY ALOUD

For material sampling without replacement, FPC multiplies the ordinary standard error and narrows the interval.

9 EXIT CHECK

Where in the CI formula does FPC belong?

ONE-GLANCE VISUAL ROUTE



8 Finite-population sample-size adjustment

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Population size N is finite.
- Initial large-population requirement n0 has been calculated.
- Need the final sample size for sampling without replacement.

2 DECISION / METHOD

1. Compute n0 using the ordinary mean/proportion formula.
2. Apply the finite-N adjustment.
3. ALWAYS ROUND UP the adjusted result.
4. Keep n0 and final n distinct.

3 FORMULA

```
n = n0*N / (n0 + (N-1))
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: this finite-population sample-size adjustment.
Still know: first calculate n0 correctly and round the adjusted n up.

5 EXCEL ROUTE

```
=ROUNDUP((n0*N)/(n0+(N-1)),0)
```

6 TINY WORKED EXAMPLE

If n0=385 and N=2000, adjusted n=322.99, so required n=323.

7 TOP EXAM TRAPS

- Applying FPC before n0 exists.
- Using current sample n where formula needs n0.
- Rounding down.
- Applying both SE-FPC and n-adjustment to the same planning step.

8 SAY ALOUD

First find the large-population n0, then shrink it for finite N and round up.

9 EXIT CHECK

Which value enters the adjustment numerator: current n or unadjusted n0?

ONE-GLANCE VISUAL ROUTE



Bootstrapping

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Small sample and population distribution unknown/non-normal.
- Need a CI for a statistic without relying on the standard normal/t route.
- A resampling table or index is supplied.

2 DECISION / METHOD

1. Start with the observed sample.
2. Resample n values FROM THAT SAMPLE WITH replacement.
3. Compute the statistic for every resample.
4. Repeat many times and order the statistics.
5. Use the cutoff convention stated/supported by the course or question.

3 FORMULA

`bootstrap CI = lower and upper empirical quantiles of resampled statistics`

4 FORMULA SHEET VS WHAT I KNOW

Gives: no bootstrap formula.
Still know: resampling source, replacement, same n, many repetitions, statistic and tail cutoffs.

5 EXCEL ROUTE

- Go to the named Bootstrap worksheet and identify the original sample/index grid.
- One mean per resample: `=AVERAGE(resample_range)`.
- Order/read the supplied resample means; use stated cutoffs.
- Preserve precision; enter requested limits to four decimals.

6 TINY WORKED EXAMPLE

Observed sample {2,4,7}. One valid n=3 resample is {4,4,2}; its bootstrap mean is 3.333. Repeat to form the bootstrap distribution.

7 TOP EXAM TRAPS

- Resampling from the population/raw database rather than the observed sample.
- Sampling without replacement or changing n.
- Treating one resample as the CI.
- Inventing a universal 250-resample cutoff-index rule.

8 SAY ALOUD

Bootstrap by repeatedly sampling with replacement from the observed sample, then use the empirical statistic distribution.

9 EXIT CHECK

Where do bootstrap draws come from, and are they drawn with replacement?

2026 A1S2 WORKBOOK WORKFLOW

1. Named Bootstrap sheet	2. Find original sample	3. Read supplied index/grid
4. Resample WITH replacement	5. One mean per resample	6. Order resample means
7. Use stated cutoff convention	8. No universal 250 rule	9. Enter 4 decimals

8 CI interpretation/reporting + method-selection tree

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need to choose among Z, t, proportion, FPC or bootstrap.
- Asked to interpret/report a finished interval.
- Closest competitor depends on parameter, σ status, normality and finite sampling.

2 DECISION / METHOD

Tree:
1. Parameter: mean or proportion?
2. Mean: σ known → Z; σ unknown → t.
3. Proportion → Z CI using p.
4. If approximate normality is doubtful / the population is clearly non-normal, use bootstrap where the course/question supports it; small n makes the normality issue especially important.
5. Finite without replacement and n/N>.05 → multiply SE by FPC.

3 FORMULA

```
report: estimate, lower, upper,
confidence level, n, units/context
CI describes parameter uncertainty,
not individual observations
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: method-specific SEs and sample-size formulas.
Still know: selection tree and correct repeated-sampling interpretation.

5 EXCEL ROUTE

Use the chosen route:
- Z: NORM.S.INV
- t: T.INV.2T, STDEV.S, df=n-1
- proportion: SQRT(p*(1-p)/n)
- FPC: SQRT((N-n)/(N-1))
- bootstrap: resample and empirical cutoffs.

6 TINY WORKED EXAMPLE

A 95% CI for μ is (12.4,15.1). Say: 'We are 95% confident the population mean lies between 12.4 and 15.1 units.'

7 TOP EXAM TRAPS

- '95% of data lie in the CI.'
- 'There is a 95% probability fixed μ is in this computed interval.'
- Using t for a proportion.
- Forgetting FPC/normality checks.

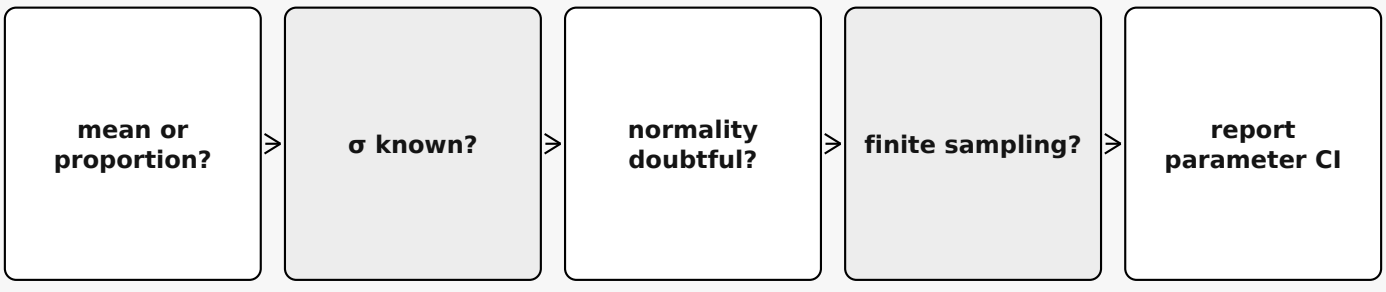
8 SAY ALOUD

Choose by parameter and σ status, then interpret the interval as uncertainty about the population parameter.

9 EXIT CHECK

Unknown σ, small normal sample, mean parameter: which distribution and df?

ONE-GLANCE VISUAL ROUTE



H_0 / H_1 + hypothesis-testing foundations

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- A claim about a population mean is tested using sample evidence.
- Need to state H_0 and H_1, a test statistic and a decision.
- Equality belongs in H_0.

2 DECISION / METHOD

1. Define the population parameter.
2. Put the benchmark/equality in H_0.
3. Let wording determine H_1: different, less, or greater.
4. Assume H_0 to build the reference distribution.
5. Quantify how surprising the sample is.

3 FORMULA

$H_0: \mu = \mu_0$
$H_1: \mu \neq \mu_0, \mu < \mu_0, \text{ or } \mu > \mu_0$
test statistic = (estimate-null)/SE_under_H0

4 FORMULA SHEET VS WHAT I KNOW

Gives: inference SE/distribution lanes.
Still know: hypothesis writing, equality placement, claim direction and contextual conclusion.

5 EXCEL ROUTE

Use cells for μ_0, α, n, $\bar{X}$ and σ/s. Excel computes the statistic; hypotheses must still be written from context.

6 TINY WORKED EXAMPLE

Claim: mean differs from 50. Write $H_0: \mu = 50$ vs $H_1: \mu \neq 50$; this is two-tail.

7 TOP EXAM TRAPS

- Putting equality in H_1.
- Writing hypotheses about $\bar{X}$ instead of population μ.
- Saying H_0 is proved.
- Choosing direction from the sample result rather than the claim.

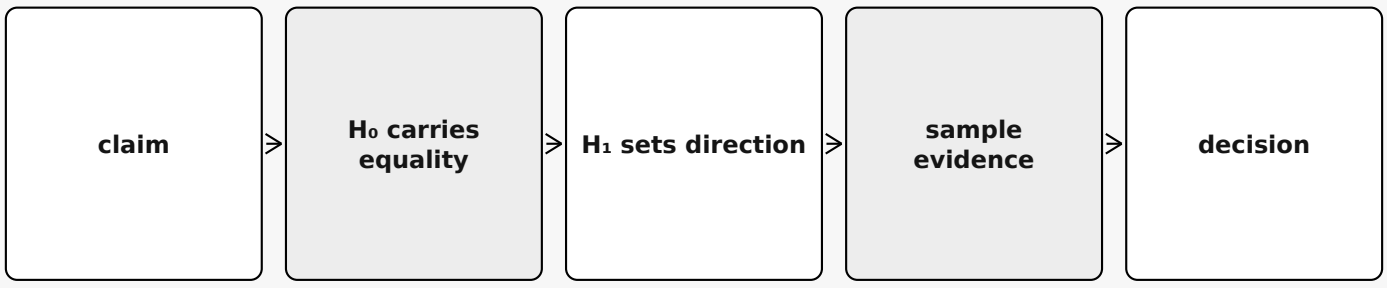
8 SAY ALOUD

H_0 carries the equality; H_1 carries the claim direction; sample evidence tells me whether to reject H_0.

9 EXIT CHECK

For the claim 'mean is below 12', what are H_0 and H_1?

ONE-GLANCE VISUAL ROUTE



9

Type I / Type II + α / β / basic power

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Asked to classify a wrong decision or compare α, β and power.
- Need qualitative effects on β and power.
- Truth of H_0 and the decision are both specified.

2 DECISION / METHOD

Use a 2x2 truth/decision grid:
- Reject true H_0 -> Type I.
- Do not reject false H_0 -> Type II.
- Correct rejection of false H_0 has probability power.

3 FORMULA

```
P(Type I)=α
P(Type II)=β
power=1-β
matching confidence coefficient=1-α
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: no error-definition formulas.
Still know - ALL ELSE FIXED: β rises when the true effect is smaller, α falls, variance rises, or n falls; power moves opposite β.

5 EXCEL ROUTE

No required calculation route here. A decision label can use:

```
=IF(p_value<alpha,"Reject H0","Do not reject H0")
```

6 TINY WORKED EXAMPLE

A real effect exists, but the test does not reject H_0: that is Type II. If β=.20, basic power is .80.

7 TOP EXAM TRAPS

- Type I = 'accepting false H_0' (wrong).
- Saying α+β=1.
- Confusing matching confidence coefficient 1-α with power 1-β.
- Omitting the ALL ELSE FIXED qualification.

8 SAY ALOUD

Type I rejects a true H_0; Type II misses a false H_0; power is the chance of detecting a false H_0.

9 EXIT CHECK

If n increases with all else fixed, what generally happens to β and power?

ONE-GLANCE VISUAL ROUTE

	H_0 TRUE	H_0 FALSE
REJECT H_0	**TYPE I probability α**	**CORRECT power = 1-β**
DO NOT REJECT H_0	**CORRECT probability 1-α**	**TYPE II probability β**

Tails + significance + rejection regions

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

A1S2 SCOPE:
Tail recognition, H_1 direction, α placement and rejection-region recognition are in scope.
The later detailed one-tail mean-test workflow after slide 51 is excluded from the current A1S2 Chapter 9 scope.

2 DECISION / METHOD

- H_1: $\mu \neq \mu_0$ → two tails, α/2 each.
- H_1: $\mu < \mu_0$ → left tail, α.
- H_1: $\mu > \mu_0$ → right tail, α.

Place the boundary/boundaries and recognise the rejection region; do not extend into later workflows.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$
two-tail criticals = $\pm z^*$
left critical = $\Phi^{-1}(\alpha)$
right critical = $\Phi^{-1}(1-\alpha)$
reject H_0 only inside rejection region

4 FORMULA SHEET VS WHAT I KNOW

Gives: no critical-value map.
Still know: H_1 controls tails; significance is total rejection-region area under H_0.

5 EXCEL ROUTE

- Two-tail: `=NORM.S.INV(alpha/2)` and `=NORM.S.INV(1-alpha/2)`
- Recognition only: left NORM.S.INV(α); right NORM.S.INV(1-α).

6 TINY WORKED EXAMPLE

If H_1:$\mu<20$ and $\alpha=.05$, it is a left-tail rejection region with Z critical about -1.6449.

7 TOP EXAM TRAPS

- Splitting α incorrectly between rejection regions.
- Choosing tail from sign of zcalc rather than H_1.
- Saying 'accept H_0'.
- Using the wrong critical-value sign.

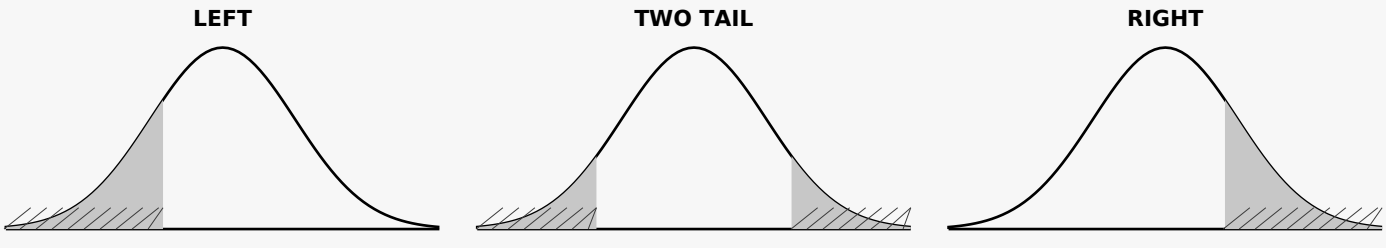
8 SAY ALOUD

The alternative hypothesis chooses the tail; α is the total rejection-region area.

9 EXIT CHECK

Which H_1 creates two rejection regions?

ONE-GLANCE VISUAL ROUTE



Mean test, σ known -> Z critical-value method

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Hypothesis test about population mean μ.
- Population σ known.
- Asked for critical value/rejection decision; here use the in-scope two-tail route.

2 DECISION / METHOD

1. State H_0/H_1 and α.
2. Check Z lane: σ known.
3. SE=σ/√n; compute zcalc.
4. Two-tail criticals use α/2.
5. Reject if zcalc is outside the criticals; conclude in context.

3 FORMULA

$Zcalc=(\bar{X}-\mu_0)/(\sigma/\sqrt{n})$

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

two-tail criticals = $\pm z^*$

reject if $|Zcalc| > z^*$

4 FORMULA SHEET VS WHAT I KNOW

Gives: Z lane and SE=σ/√n.
Still know: test statistic, critical values, decision rule and wording.

5 EXCEL ROUTE

- n=COUNT(range); Xbar=AVERAGE(range)
- criticals: NORM.S.INV(alpha/2), NORM.S.INV(1-alpha/2)
- zcalc=(X̄-μ_0)/(σ/SQRT(n)).

6 TINY WORKED EXAMPLE

H_0:μ=100, H_1:μ≠100, α=.05, X̄=104,σ=12,n=36. zcalc=2.00; |2|>1.96 -> reject H_0.

7 TOP EXAM TRAPS

- Using sample s despite supplied σ.
- Comparing X̄ directly with a Z critical.
- Using one critical for a two-tail test.
- Saying reject because z is positive.

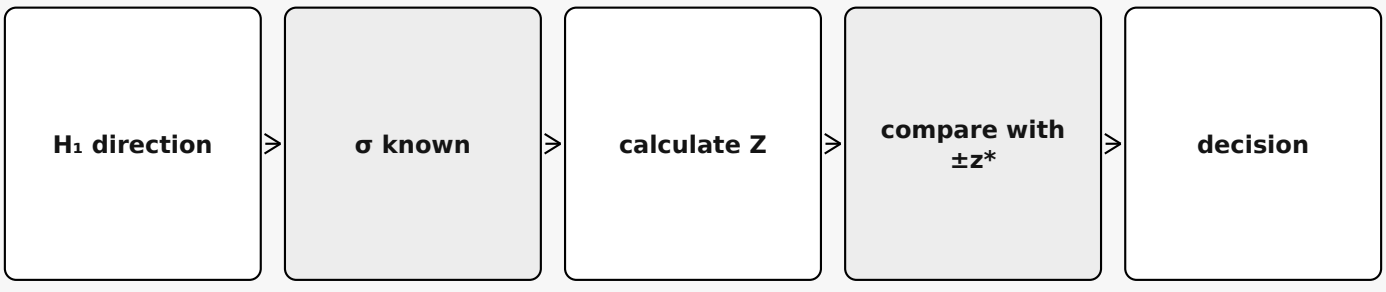
8 SAY ALOUD

Known σ means standardise X̄ with σ/√n, then compare zcalc with the rejection boundary.

9 EXIT CHECK

If zcalc=1.4 in a two-tail α=.05 Z test, what is the decision?

ONE-GLANCE VISUAL ROUTE



p-value method + Excel execution

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Asked for p-value or Excel decision for a mean test.
- This sheet focuses on the in-scope two-tail Z route.
- p-value measures extremeness assuming H_0.

2 DECISION / METHOD

1. Compute zcalc under H_0.
2. For two tails, double the upper-tail area beyond |zcalc|.
3. Compare p-value with α.
4. p<α -> reject H_0; otherwise do not reject H_0.
5. State contextual evidence.

3 FORMULA

```
two-tail p = 2*[1-Φ(|zcalc|)]
reject H0 if p-value < α
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: Z lane and SE.
Still know: p-value definition, two-tail doubling, decision comparison and conclusion language.

5 EXCEL ROUTE

- `n=COUNT(range); Xbar=AVERAGE(range)`
- `zcalc=(Xbar-mu0)/(sigma/SQRT(n))`
- `=(1-NORM.S.DIST(ABS(zcalc),TRUE))*2.`

6 TINY WORKED EXAMPLE

If zcalc=-2.10, two-tail p is about .0357. At α=.05, reject H_0.

7 TOP EXAM TRAPS

- p-value=P(H_0 true).
- Forgetting ABS or the factor 2.
- Rejecting when p>α.
- Reporting 'accept H_0'.

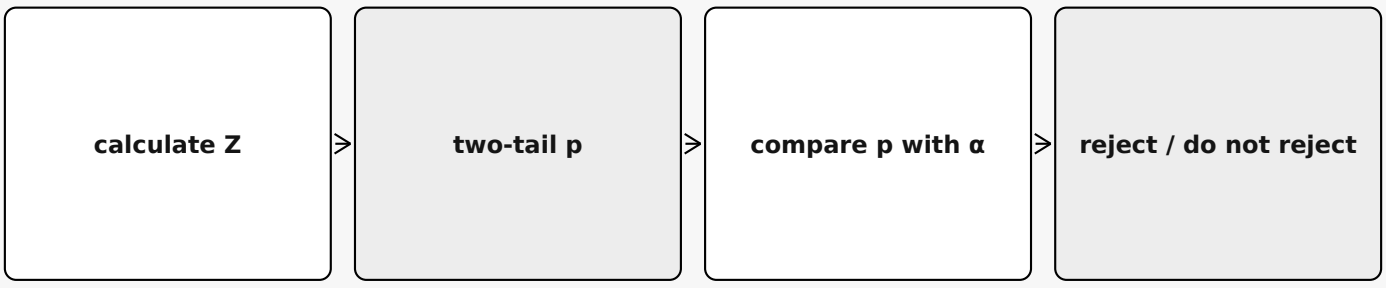
8 SAY ALOUD

The p-value is how extreme the sample is under H_0; below α means reject H_0.

9 EXIT CHECK

Write the exact Excel formula for a two-tail Z p-value from zcalc in B11.

ONE-GLANCE VISUAL ROUTE



Two-tail Z test ↔ confidence interval

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Two-tail mean test at α and matching confidence coefficient 1-α.
- Population σ known.
- Asked to compare test and CI conclusions.

2 DECISION / METHOD

1. Build the matching Z CI: $\bar{X} \pm z^*SE$.
2. If μ_0 lies outside CI -> reject H_0:$\mu=\mu_0$.
3. If μ_0 lies inside CI -> do not reject H_0.
4. This equivalence is for matching two-tail α/confidence.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$CI: \bar{X} \pm z^* \times \sigma/\sqrt{n}$

two-tail reject H_0 iff μ_0 is outside CI

4 FORMULA SHEET VS WHAT I KNOW

Gives: Z SE lane.
Still know: match confidence coefficient 1-α and distinguish a mean CI from an individual-value range.

5 EXCEL ROUTE

- z=NORM.S.INV(1-alpha/2)
- lower/upper=$\bar{X}$ ± z*σ/SQRT(n)
- =IF(OR(mu0<lower,mu0>upper),"Reject H0","Do not reject H0").

6 TINY WORKED EXAMPLE

$\bar{X}$=52,σ=8,n=64,α=.05 gives CI (50.04,53.96). μ_0=50 is outside -> reject H_0 in the matching two-tail test.

7 TOP EXAM TRAPS

- Using a 90% CI for α=.05.
- Applying equivalence to a one-tail test.
- Saying values outside CI are impossible observations.

8 SAY ALOUD

For a matching two-tail test, reject exactly when the hypothesised mean is outside the confidence interval.

9 EXIT CHECK

At α=.01, which confidence level must the comparison interval use?

ONE-GLANCE VISUAL ROUTE



Mean test, σ unknown -> Student t + normality checks

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Hypothesis about mean μ; population σ unknown.
- Raw sample or $\bar{X}$,s,n supplied.
- Need to justify the t lane and inspect normality.

2 DECISION / METHOD

1. Compute n, $\bar{X}$, s.
2. Check random sampling and approximate population normality.
3. Use tcalc with SE=s/√n.
4. df=n-1.
5. Continue with critical or p-value route.

3 FORMULA

$t_{calc}=(\bar{X}-\mu_0)/(s/\sqrt{n})$
$df=n-1$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $t_{(n-1)}$ and $s/\sqrt{n}$ when σ unknown.
Still know: normality evidence: symmetry, histogram/boxplot and linear normal probability plot.

5 EXCEL ROUTE

- n=COUNT(range)
- Xbar=AVERAGE(range)
- s=STDEV.S(range)
- tcalc=(Xbar-mu0)/(s/SQRT(n)); df=n-1.

6 TINY WORKED EXAMPLE

$\bar{X}=31.5, s=6, n=16, \mu_0=30$ -> SE=1.5, tcalc=1.0, df=15. Normality should be checked because the sample is small.

7 TOP EXAM TRAPS

- Using Z with sample s.
- df=n.
- Treating a single outlier-free boxplot as proof.
- Forgetting the population-normality condition for small n.

8 SAY ALOUD

Unknown σ sends the mean test to Student t with sample s, n-1 df, and a normality check.

9 EXIT CHECK

What three Excel summaries are needed before a raw-data t test?

ONE-GLANCE VISUAL ROUTE

μ test > σ unknown > normality check > s/√n, df=n-1 > Student t

Two-tail t critical/p-value method + t test ↔ CI

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Two-tail mean test; σ unknown; t assumptions acceptable.
- Need critical values, p-value and/or matching CI conclusion.
- This is the final in-scope Chapter 9 method family.

2 DECISION / METHOD

1. tcalc=($\bar{X}$-μ_0)/(s/√n); df=n-1.
2. Critical route: compare |tcalc| with T.INV.2T(α,df).
3. p route: T.DIST.2T(ABS(tcalc),df).
4. Reject iff p<α.
5. Matching t CI excludes μ_0 iff test rejects.

3 FORMULA

```
t* = T.INV.2T(α,df) > 0
two-tail p=T.DIST.2T(|tcalc|,df)
CI: X¯ ± t* × s/√n
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: t lane, df and SE.
Still know: t critical/p-value functions and two-tail CI equivalence.

5 EXCEL ROUTE

- `tstar=T.INV.2T(alpha,df)`
- `p=T.DIST.2T(ABS(tcalc),df)`
- CI endpoints=`Xbar +/- tstar*s/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=18.4,s=2.5,n=25,μ_0=17: tcalc=2.8,df=24,p about .0099. At α=.05 reject; matching CI (17.368,19.432) excludes 17.

7 TOP EXAM TRAPS

- Using T.DIST instead of T.DIST.2T for a two-tail p-value.
- Using NORM.S.INV for t critical.
- Comparing μ_0 with a prediction range instead of mean CI.

8 SAY ALOUD

For a two-tail t test, T.DIST.2T gives the p-value and the matching t interval gives the same decision.

9 EXIT CHECK

Which Excel function returns the two-tail t p-value?

ONE-GLANCE VISUAL ROUTE

